Case 03 · Specification: Roll Transparency Log

Following chapter 2 (specification) of the Magshimim project-book template: the product promise, the features by layer, and for each feature its purpose, steps, record, threat and answer; then technologies, decisions, what changed since the POC, and what stays open. Characters are fictional.

Opening scenario

On the morning the draft roll was published, Priya, a nurse in Lucknow, learned from a friend's screenshot that her name was gone from booth 212 in her parents' village. The reason read "absent, shifted or other". Nobody called. She does not know who removed her, when, or why. More than 130 million names left the draft rolls in a process run centrally, with software that did not show who was removed and why. The system specified here is a transparency log for the roll: every change is recorded with a reason, cannot be altered quietly, and every person can check themselves.

What the product promises

- Every change to the roll, with its reason, is kept so that it cannot be altered quietly.
- Every person checks their own line in a second, and receives a proof.
- No names on the log, and no need to trust whoever runs it.

Structure: four layers, ten features

Layer	Features
Intake	1. Change capture · 2. Detailed reason codes · 3. Keyed identifiers
The log	4. Merkle log · 5. Signed tree heads · 6. Roll-to-log reconciliation
At the voter	7. Self-check · 8. Alert and appeal
Witnesses	9. Witness network · 10. Aggregate data

Layer 1: Intake

Feature 1: Change capture

Every officer action is sent to the log before it takes effect. Change data capture (Debezium) reads each new row from the existing roll database's write-ahead log and publishes a ChangeEvent to Kafka, partitioned by district, retained for 14 days. The officer terminal does not show "saved" until the log returns a receipt.

Threat: a change that bypasses the log. **Answer:** the roll is rebuilt from the log nightly (feature 6); any change without a log line is flagged.

Feature 2: Detailed reason codes

Every deletion says why, from a closed list of 11 codes, with evidence. A-NEW, A-RESTORE, R-DEAD, R-SHIFT, R-ABSENT, R-DUP, R-NOFORM, R-REQ, R-COURT, M-MOVE, M-FIX. There is no "other". R-DEAD requires a hash of a death record; R-SHIFT requires the new address or a field note.

Threat: an officer picks a convenient code. **Answer:** evidence is mandatory, and the person sees the code the same day.

Feature 3: Keyed identifiers

No names or numbers on the log. Voters and officers appear as HMAC-SHA256 pseudonyms whose key never leaves an HSM.

Threat: a dictionary of all elector numbers. **Answer:** without the key, no dictionary can be built.

Layer 2: The log

Feature 4: Merkle log

Every line is a leaf in an append-only Merkle tree (RFC 6962/9162), run on Tessera and served as static tiles from a CDN. A correction is a new M-FIX line, never an edit.

Threat: the operator rewrites an old line and recomputes the tree. **Answer:** the new head no longer extends the old one; anyone holding the old head knows (caught in the POC).

Feature 5: Signed tree heads

Every minute: size and root, signed by the Commission in an HSM (Ed25519), then cosigned by at least three of five witnesses (C2SP signed-note).

Feature 6: Roll-to-log reconciliation

The published roll must equal the sum of all log changes, booth by booth. A per-booth Merkle root is compared nightly and at every publication.

Layer 3: At the voter

Feature 7: Self-check

Priya sends her elector number and date of birth and gets her status, history, reason in plain words, and the last date to appeal, with an inclusion proof her phone verifies against a cosigned head. Channels: PWA, SMS, booth kiosk. Rate-limited.

Feature 8: Alert and appeal

Whoever left a phone number gets an SMS the day their line changes, with a link to a pre-filled appeal (Form 6) carrying the line's identifier.

Layer 4: Witnesses

Feature 9: Witness network

Newspapers, NGOs, parties and universities run witnesses (tlog-witness) that check every minute that the log only grows.

Feature 10: Aggregate data

Per district, week and code: how many added, removed, restored. Cells below 10 are suppressed. No booth, no date, no names.

Key technologies

Technology	Role	Why
Debezium + Kafka	Change capture and durable queue	No change to the Commission's existing software

Tessera	Tile-based transparency log (successor to Trillian)	Static tiles survive draft-day load
RFC 6962/9162	Leaf/node hashing, inclusion and consistency proofs	Any CT-aware tool can verify the log
HSM, HMAC-SHA256, Ed25519	Pseudonyms and signatures	Keys never leave hardware
tlog-witness, C2SP signed-note	Witness cosigning	No need to trust the operator
PWA, SMS gateway	Self-check and alerts	Reaches voters without smartphones too

Decisions that could have gone the other way

- **Who runs the log?** The Commission, with witnesses. Only it can write to the roll in time; trust comes from who approves, not who runs. *Price: dependence on its availability.*
- **Plain or keyed identifier?** Keyed, in an HSM. *Price: an HSM to operate.*
- **When is a change logged?** Before it takes effect. *Price: the log becomes a dependency of every change.*
- **How many witness signatures?** Three of five. *Price: coordination with outside bodies.*
- **What does everyone see?** Aggregates only. *Price: less detail for researchers.*
- **Keep "other"?** No. *Price: officers must choose.*

What changed since the POC

In the POC	In the spec
HMAC key inside the process	HMAC key inside an HSM
Booth and date on every public line	Aggregate public feed; full detail only in the personal proof
Signature modelled with HMAC	Ed25519, plus three witness cosignatures
No check that a change was sent	Roll-to-log reconciliation, booth by booth
One imagined monitor	Five or more witnesses, three signatures
Python in one process	Tessera, tiles on a CDN, website, SMS and kiosk

What the spec closed, and what stays open

Closed: silent rewrite, split view, a change that skips the log, a voter who doesn't know, and the opaque reason bucket. The core rests on the Certificate Transparency standard and on code already in production.

Open: the log proves what was recorded, not that it was right. If an officer records R-DEAD for a living person, the log keeps it faithfully. What changes is that the living person can see it, in time, and appeal. Who hears the appeal in time? The log does not solve that.